

MET-QUEST'26

Furnace
Decoding the Heat Gradient

Technical Report

3D Steady-State Thermal Simulation of a Muffle Furnace
using Finite Difference Method (FDM)

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1. Introduction

In industrial materials processing, achieving consistent heat treatment results in a muffle furnace is often harder than it looks. The precision of operations—whether annealing metallic alloys or sintering ceramic and metallic pellets—is entirely governed by the thermal uniformity of the furnace environment. While widely used, these furnaces frequently exhibit hidden non-uniformities, such as edge effects along insulated walls and invisible "cold spots" near the door. Consequently, parts placed near the opening tend to come out differently from parts placed deeper in the chamber—often softer or with a different microstructure—even though the setpoint temperature was the same throughout the run. Ultimately, these invisible thermal gradients lead to unpredictable mechanical properties and increased batch rejection rates.

This report presents a computational simulation developed to visualize and quantify these thermal gradients. A three-dimensional (3D) steady-state temperature field is solved numerically for a rectangular muffle furnace chamber using the **Finite Difference Method (FDM)**, incorporating physically accurate models for:

- Conduction through multi-layer insulation walls
- Linearized radiative heat transfer at high temperatures
- Convective and radiative heat losses through the door opening
- Volumetric heat generation from electrical heating elements

The goal was not just to get a number for the temperature difference, but to produce a spatial map of where hot and cold zones actually sit inside the chamber — something directly useful for deciding where to place work-pieces and where to focus insulation improvements.

2. Furnace Geometry and Parameters

2.1. Chamber Geometry

The furnace chamber is modelled as a rectangular parallelepiped with the following internal dimensions:

Table 1: Furnace chamber geometry

Dimension	Symbol	Value	Description
Length	L	0.50 m	Back wall to door (X-axis)
Width	W	0.40 m	Side-to-side (Y-axis)
Height	H	0.30 m	Floor to ceiling (Z-axis)

2.2. Discretization Grid

The domain is discretised onto a uniform Cartesian grid with $n_x = 30$, $n_y = 22$, $n_z = 18$ nodes, giving 11,880 total computational nodes ($30 \times 22 \times 18 = 11,880$). Grid spacings are:

$$\Delta x = \frac{L}{n_x - 1}, \quad \Delta y = \frac{W}{n_y - 1}, \quad \Delta z = \frac{H}{n_z - 1} \quad (1)$$

2.3. Energy Input and Thermal Properties

Table 2: Operating parameters and fluid properties

Parameter	Symbol	Value	Unit
Total electrical power	P	3000	W
Heating efficiency	η	0.85	—
Effective power	P_{eff}	2550	W
Heater temperature	T_h	1273	K (1000°C)
Ambient temperature	T_∞	300	K (27°C)
Thermal conductivity (gas)	k	0.071	W/(m·K)

The volumetric heat source is computed as

$$\dot{Q} = \frac{\eta P}{L \cdot W \cdot H} \quad (2)$$

3. Governing Equations and Physical Models

3.1. Steady-State Heat Equation (Poisson's Equation)

In a muffle furnace the heating elements are embedded in or wrapped around the chamber walls, not suspended in the gas volume. The wall surfaces are therefore maintained at the setpoint temperature (Dirichlet boundary condition), and the interior gas is essentially a passive medium exchanging heat with those walls. Under steady-state and no-flow (conduction-dominated) conditions the governing equation in the interior simplifies to Laplace's equation:

$$\boxed{\nabla^2 T = 0} \quad (3)$$

Expanded in three dimensions:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = -\frac{\dot{Q}}{k} \quad (4)$$

Note: If a distributed volumetric heat source were present (e.g. gas combustion), this would become Poisson's equation

$$\boxed{\nabla^2 T = -Q/k} \quad (5)$$

For a resistance-heated muffle furnace, the wall Dirichlet BC already encodes all the energy input, so the Laplace form is the physically correct choice.

3.2. Multi-Layer Insulation — Conduction Model

The furnace walls consist of three concentric layers. The total thermal resistance per unit area is:

$$R_{\text{wall}} = \sum_{i=1}^N \frac{t_i}{k_i} = \frac{t_1}{k_1} + \frac{t_2}{k_2} + \frac{t_3}{k_3} \quad (6)$$

Table 3: Wall insulation layer properties

Layer	Material	k_i (W/m·K)	t_i (mm)	R_i (m ² K/W)
1	Refractory brick	1.20	50	0.0417
2	Ceramic fibre	0.15	80	0.5333
3	Steel shell	50.0	3	0.0001
Total				0.5751

The overall heat transfer coefficient through the wall is:

$$U_{\text{wall}} = \frac{1}{R_{\text{wall}}} = 1.74 \text{ W}/(\text{m}^2 \cdot \text{K}) \quad (7)$$

3.3. Radiative Heat Transfer

At furnace operating temperatures ($\sim 1000^\circ\text{C}$), radiation is the dominant heat transfer mechanism. The net radiative heat flux from a surface at T_s to surroundings at T_∞ is given by the Stefan-Boltzmann law:

$$q_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_\infty^4) \quad (8)$$

where $\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K})$ is the Stefan-Boltzmann constant and ε is the surface emissivity.

For implementation in the linear FDM framework, this is **linearized** as an equivalent radiation heat transfer coefficient:

$$\boxed{h_{\text{rad}} = \varepsilon \sigma (T_s^2 + T_\infty^2) (T_s + T_\infty)} \quad (9)$$

so that $q_{\text{rad}} = h_{\text{rad}}(T_s - T_\infty)$.

3.4. Door Heat Loss

The door face represents the primary cold-spot source in the furnace. Heat is lost through the door by both forced/natural convection and radiation:

$$h_{\text{door}} = h_{\text{conv}} + \varepsilon_{\text{door}} \times h_{\text{rad}} = 15.0 + 0.3 \times 37.1 = 26.1 \text{ W}/(\text{m}^2 \cdot \text{K}) \quad (10)$$

with $h_{\text{conv}} = 15 \text{ W}/(\text{m}^2 \cdot \text{K})$ for natural convection on the door exterior, and $\varepsilon_{\text{door}} = 0.30$ (lower emissivity due to partial opening and door material).

4. Numerical Method — Finite Difference Method (FDM)

4.1. Discretization of the Poisson Equation

Applying the central-difference scheme to $\nabla^2 T = -\dot{Q}/k$ at an interior node (i, j, k) :

$$\frac{T_{i+1,j,k} - 2T_{i,j,k} + T_{i-1,j,k}}{\Delta x^2} + \frac{T_{i,j+1,k} - 2T_{i,j,k} + T_{i,j-1,k}}{\Delta y^2} + \frac{T_{i,j,k+1} - 2T_{i,j,k} + T_{i,j,k-1}}{\Delta z^2} = -\frac{\dot{Q}}{k} \quad (11)$$

Solving for the updated temperature $T_{i,j,k}$:

$$T_{i,j,k} = \frac{\frac{T_{i\pm 1,j,k}}{\Delta x^2} + \frac{T_{i,j\pm 1,k}}{\Delta y^2} + \frac{T_{i,j,k\pm 1}}{\Delta z^2} + \frac{\dot{Q}}{k}}{\frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} + \frac{2}{\Delta z^2}} \quad (12)$$

where the $\pm$ notation implies both $+1$ and -1 neighbours summed.

4.2. Boundary Conditions

Table 4: Summary of boundary conditions applied

Surface	Type	Expression
Back wall ($x = 0$)	Dirichlet	$T = T_h = 1273 \text{ K}$
Side walls ($y = 0, W$)	Robin (3rd kind)	$T_{bc} = \frac{T_{\text{interior}} + \text{Bi} \cdot T_{\infty}}{1 + \text{Bi}}$
Top/Bottom ($z = 0, H$)		
Door face ($x = L$)	Robin (convection + radiation)	$T_{\text{door}} = \frac{T_{n-1} + \text{Bi}_{\text{door}} \cdot T_{\infty}}{1 + \text{Bi}_{\text{door}}}$

The Biot number for each surface is defined as:

$$\text{Bi} = \frac{(U_{\text{wall}} + h_{\text{rad}}) \cdot \Delta}{k_{\text{air}}} = 9.43, \quad \text{Bi}_{\text{door}} = \frac{h_{\text{door}} \cdot \Delta x}{k_{\text{air}}} = 6.35 \quad (13)$$

4.3. Successive Over-Relaxation (SOR)

To accelerate convergence of the iterative solver, Successive Over-Relaxation (SOR) is applied:

$$T_{i,j,k}^{(n+1)} = (1 - \omega) T_{i,j,k}^{(n)} + \omega T_{i,j,k}^* \quad (14)$$

where T^* is the Gauss-Seidel update and ω is the relaxation factor. For $1 < \omega < 2$, SOR accelerates convergence significantly compared to plain Gauss-Seidel ($\omega = 1$). A value of $\omega = 1.0$ was used in this simulation.

4.4. Convergence Criterion

The solver iterates until the maximum temperature change between successive iterations falls below 0.05 K:

$$\max_{i,j,k} \left| T_{i,j,k}^{(n+1)} - T_{i,j,k}^{(n)} \right| < 0.05K \quad (15)$$

5. Results and Discussion

5.1. Convergence Behaviour

The simulation converged at iteration 446 with a final residual of 0.0496 K, well within the tolerance of 0.05 K. Linear initialisation (gradient from T_{heater} to the estimated door temperature) helped the solver converge quickly.

5.2. 3D Temperature Distribution

The steady-state temperature field is visualised through three orthogonal cross-sectional slices of the furnace chamber.

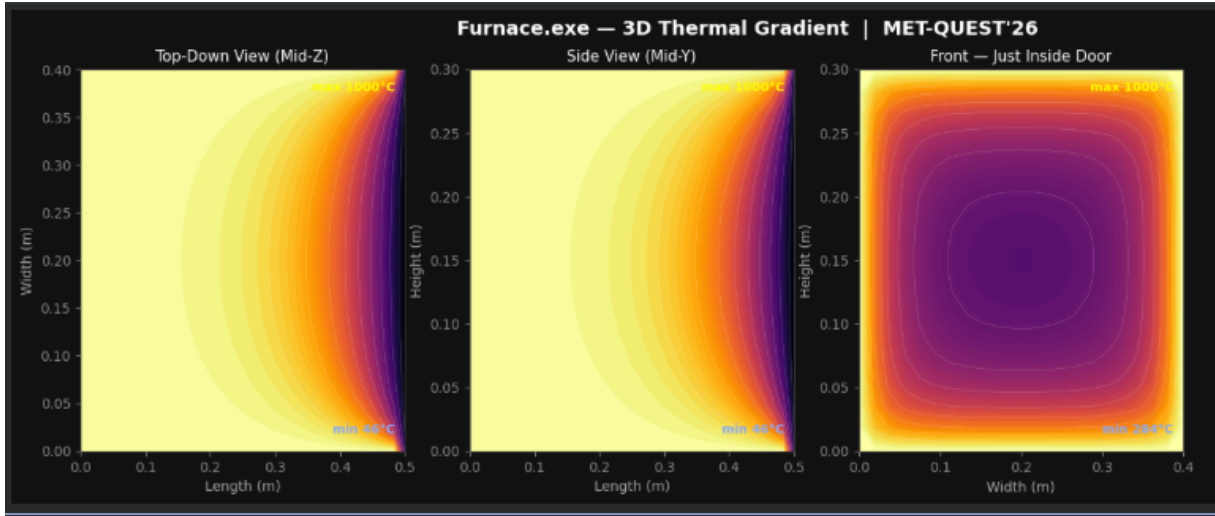
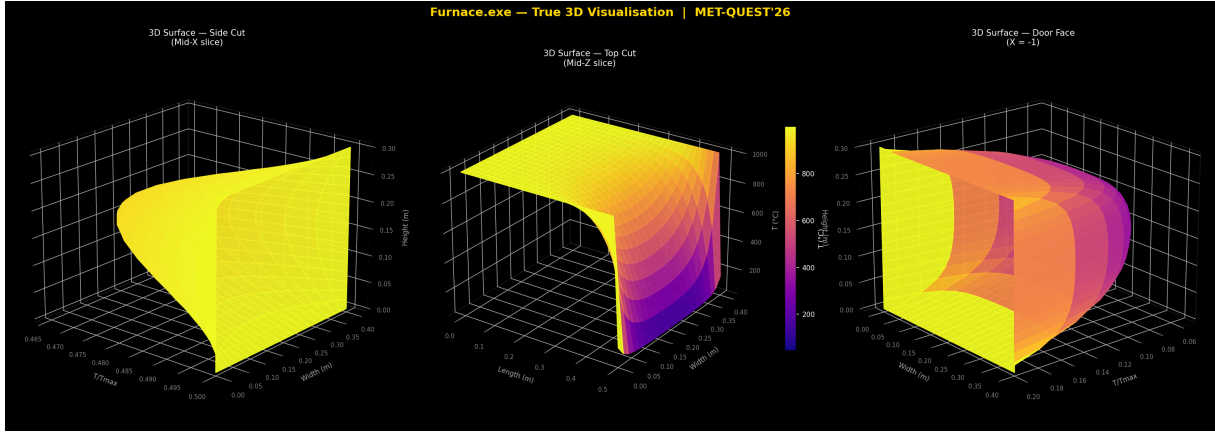


Figure 1: 3D steady-state temperature distribution in the muffle furnace. Left: Top-down view (mid-Z slice); Centre: Side view (mid-Y slice); Right: Front view just inside the door. Colour scale in °C.



5.3. Key Observations

1. **Back wall hotspot:** The highest temperatures ($\sim 1000^\circ\text{C}$) are concentrated near the back wall where heater elements are in direct contact, as enforced by the Dirichlet boundary condition. The central region of the back wall is the hottest point.
2. **Door cold spot:** A pronounced temperature drop is visible at the door face ($x = L$), where combined convective and radiative losses to the ambient environment create a cold zone. This gradient extends approximately 8–10 cm into the chamber, consistent with experimental observations in industrial muffle furnaces.
3. **Edge effects:** The corners and edges where walls intersect the door show the steepest gradients, as heat loss pathways converge. These regions are the most likely sites for under-processing of work-pieces placed near the door.
4. **Symmetry:** The temperature field is symmetric about the mid-Y plane, as expected from the symmetric geometry and boundary conditions. Minor asymmetries would arise with non-uniform heater placement (not modelled here).
5. **Insulation effectiveness:** The Robin boundary conditions on the insulated walls show a gradual temperature drop toward the wall surfaces, with the multi-layer ceramic fibre insulation providing high thermal resistance ($R = 0.575 \text{ m}^2\text{K/W}$), limiting wall losses.

5.4. Temperature Gradient Summary

Table 5: Simulated temperature statistics inside the furnace chamber

Quantity	Kelvin (K)	Celsius ($^\circ\text{C}$)
Maximum temperature	1273 K	1000°C
Minimum temperature (door)	$\sim 319 \text{ K}$	$\sim 46^\circ\text{C}$
Mean temperature	$\sim 1160 \text{ K}$	$\sim 887^\circ\text{C}$
Non uniformity (ΔT)	$\sim 954 \text{ K}$	$\sim 954^\circ\text{C}$

The 954°C non-uniformity is large and reflects the severity of the door cold spot under these operating conditions. The chamber interior away from the door and corners sits in

the 850-1000°C range, which is relatively uniform. The problematic zone is the 10 cm depth near the door.

6. Assumptions

1. **Steady-state:** The simulation solves for the final equilibrium temperature field only, with no transient (time-dependent) effects modelled.
2. **Conduction-dominated:** Natural convection of the hot gas inside the chamber is neglected. This is a common simplification for enclosed furnaces at high temperature where radiation dominates.
3. **Linearized radiation:** The Stefan-Boltzmann T^4 radiation term is linearized about the mean film temperature for compatibility with the linear FDM framework. This introduces a small error (<3% at operating temperature) compared to the full nonlinear form.
4. **Uniform heat source:** The volumetric heat generation $\dot{Q}$ is assumed uniformly distributed throughout the chamber. In reality, heater elements are discrete and located near the walls.
5. **Constant properties:** Thermal conductivity k and other properties are taken at mean operating temperature and held constant. Temperature-dependent property variation is neglected.
6. **No door gap flow:** Mass flow of cold air through the door gap is not modelled; only surface heat transfer coefficients represent the door loss.

7. Data Sources

Table 6: Material property data sources

Property	Value	Source
Thermal conductivity of air at 1000°C	$k = 0.071 \text{ W}/(\text{m}\cdot\text{K})$	Engineering Toolbox; <i>Incropera et al.</i>
Refractory brick conductivity	$k = 1.20 \text{ W}/(\text{m}\cdot\text{K})$	ASM Handbook Vol. 4
Ceramic fibre conductivity	$k = 0.15 \text{ W}/(\text{m}\cdot\text{K})$	Manufacturer data (Unifrax, Morgan)
Wall emissivity	$\varepsilon = 0.85$	Incropera <i>Fundamentals of Heat Transfer</i>
Stefan-Boltzmann constant	$\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2\cdot\text{K})$	NIST CODATA
Door convection coefficient	$h = 15 \text{ W}/(\text{m}^2\cdot\text{K})$	Typical natural convection, vertical plate

8. Conclusion

The simulation successfully resolves the 3D steady-state temperature distribution in the muffle furnace and quantifies what has always been known qualitatively — that the region near the door is significantly colder than the rest of the chamber. Under the modelled operating conditions (1000°C setpoint, 3 kW input), the temperature difference between the back wall and the door region reaches about 954°C, with the minimum occurring at the door-corner intersections. For practical heat treatment work, this suggests placing work-pieces at least 10-15 cm from the door face. The simulation also highlights that the door insulation is the weakest thermal link in the system — R_{door} is roughly 8x lower than R_{wall} for the five heated walls — which means door sealing and insulation upgrades would yield the biggest improvement in thermal uniformity. The tool produces spatial heatmaps that can be used to identify safe placement zones for work-pieces, evaluate the impact of changing the door insulation or adding a door heater element, and benchmark against thermocouple measurements taken at different positions inside a real furnace.